

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Velan R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

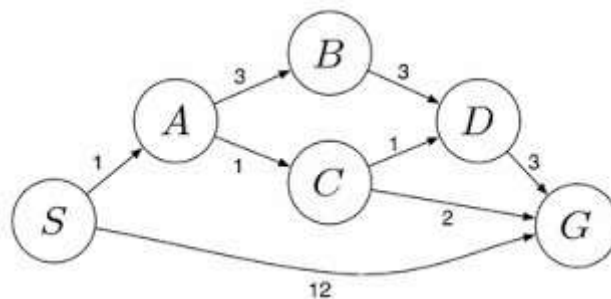
Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit

assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent? ii. Characterize the operating environment. iii. What are the actions the agent can take? iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ANSWERS

1. Water jug Problem

Given:

Jug A = 4 Gallons

Jug B = 3 Gallons

The goal is to measure exactly 2 gallons in 4 gallons jug

Steps:

Step 1 – Fill the 3 gallons jug fully.

Step 2 – Pour that 3 gallons into the 4 gallons jug.

Step 3 – Now fill the 3 gallons jug again.

Step 4 – Pour from the 3 gallons into the 4 gallons until the 4 gallons jug is full now the 3 gallons jug will be having exactly 2 gallons.

Step 5 – Now empty the 4 gallons jug.

Step 6 – Pour the 3 gallons jug into the 4 gallons.

RESULT:

The 4 gallons jug will be having 2 gallons now.

2. Mars Rover Agent

i. What are the percepts for this agent?

The Mars Rover perceives:

- Images from cameras
- Rock and soil composition
- Temperature
- Atmospheric pressure
- Terrain obstacles
- Distance to rocks
- Position (GPS/Inertial sensors)
- Battery level
- Wheel condition

ii. Characterize the operating environment

Property	Description
Observability	Partially Observable
Agents	Single Agent
Deterministic/Stochastic	Stochastic
Episodes	Sequential
Dynamics	Dynamic
State	Continuous
Knowledge	Known but uncertain

iii. Actions the agent can take

- Move forward/backward
- Turn left/right
- Capture images
- Collect rock samples
- Drill rocks
- Analyze soil
- Send data to Earth
- Avoid obstacles

- Recharge using solar panels

iv. Performance Measure

The rover is evaluated by:

- Number of successful samples collected
- Accuracy of scientific analysis
- Distance explored
- Battery efficiency
- Obstacle avoidance
- Mission completion time
- Safe operation

v. Suitable Agent Architecture

Goal-Based Agent

Why?

- It plans paths toward scientific targets.
- Chooses actions to achieve mission goals.
- Can adapt when obstacles appear.
- Makes intelligent decisions instead of fixed responses.

3.8 Queens Problem

Initial State

8 queens placed randomly (or empty chessboard).

State Space

All possible arrangements of 8 queens.

Actions

Move a queen within its column to another row.

Goal Test

No queen attacks another.

- No same row
- No same column
- No same diagonal

Path Cost

Each move costs 1.

Search Strategy

Use:

- Backtracking Search
- Depth First Search (DFS)

Algorithm

1. Place queen in first column.
2. Check if position is safe.
3. If safe, place next queen.
4. If not safe, try another row.
5. If no row works, backtrack.
6. Continue until all 8 queens are placed.

Solution State

A valid solution satisfies:

- No two queens share a row.
- No two queens share a column.
- No two queens share a diagonal.

4. Cab Booking Mobile Application

Suitable Agent

Utility-Based Agent

Why?

The customer wants the **best option** considering:

- Fare
- Comfort
- Waiting time

- Travel time

The utility-based agent compares all options and chooses the one with maximum utility.

Pseudocode:

START

Input source

Input destination

Display available cabs

FOR each cab

Calculate fare

Calculate waiting time

Calculate comfort score

Compute utility

END FOR

Select cab with highest utility

Book cab

Display booking confirmation

END

5. Uniform Cost Search (UCS)

The graph is **not visible** in the uploaded image (it is cut off below the text), so the exact UCS solution cannot be computed.

UCS Algorithm

1. Start at source node **S**.
2. Add **S** to the priority queue with cost 0.
3. Remove the node with the lowest path cost.
4. If it is the goal **G**, stop.
5. Expand its neighbors.
6. Add neighbors with cumulative costs.
7. Repeat until **G** is reached.

From the given graph:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Path Cost Calculation

Path 1

$$S \rightarrow G$$

$$\text{Cost} = 12$$

Path 2

$$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$$

$$= 1 + 3 + 3 + 3$$

$$= 10$$

Path 3

$$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$$

$$= 1 + 1 + 1 + 3$$

$$= 6$$

Path 4

$$S \rightarrow A \rightarrow C \rightarrow G$$

$$= 1 + 1 + 2$$

$$= 4 \text{ (Minimum Cost)}$$

Final Answer

- **Least-Cost Path:** $S \rightarrow A \rightarrow C \rightarrow G$
- **Total Cost:** 4

Pseudocode:

UCS(Graph, Start, Goal)

PriorityQueue $\leftarrow \{(Start,0)\}$

Visited $\leftarrow \{\}$

while PriorityQueue not empty

 Remove node with lowest cost

 if node = Goal

 return path

 Mark node visited

 for each neighbor

 if neighbor not visited

 Insert neighbor with cumulative cost

return Failure